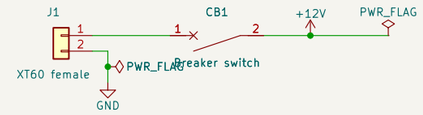
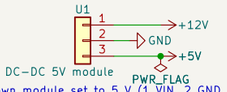
Power input and breaker switch



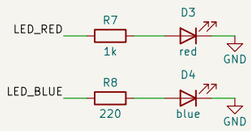
J1: female XT60 on the case, for the battery (LiFePO4 12.8 V 6 Ah with BMS) or a lab supply (12–14 V, current limit about 2 A).
The battery is charged off the mount.
CB1: resettable breaker switch, 3–5 A, rated for DC.

5 V supply for the ESP32 board



U1: DC–DC step-down module set to 5 V (1 VIN, 2 GND, 3 VOUT), off the board, wired to pads P5.
CB1 on: keep the ESP32 board's USB unplugged.
USB plugged in: keep CB1 off. The board's USB and 5V pin are not meant to supply power together.

LEDs (on the case)



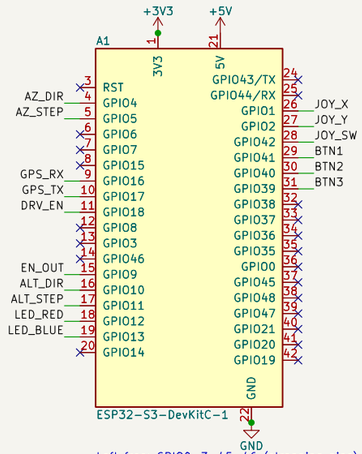
GPIO high = LED on.
PWM sets the brightness.

GPS (on the case)



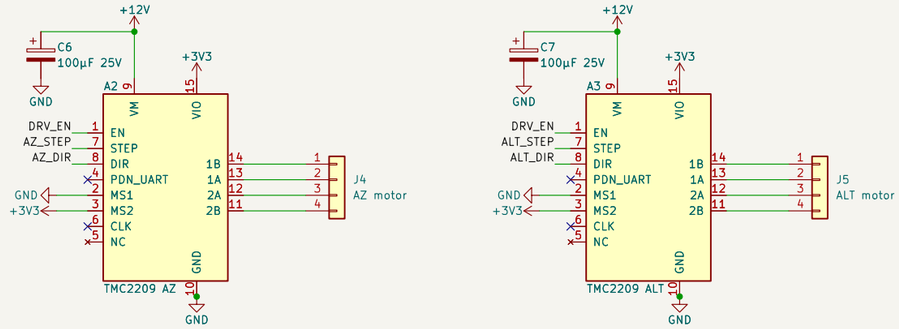
GPS module (u-blox NEO-6M): 1 VCC, 2 RX, 3 TX, 4 GND. VCC 3.3–5 V, 3.3 V logic, UART 9600 baud. Antenna towards the sky. Wired to pads P4.

Controller



Left free: GPIO0, 3, 45, 46 (strapping pins), GPIO19/20 (USB), GPIO35–37 (PSRAM on some boards), GPIO38 (RGB LED), GPIO43/44 (USB serial console).

Stepper drivers



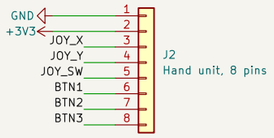
Standalone mode, no UART: MS1 = GND, MS2 = 3V3 gives 1/64 microsteps.
Motor current: potentiometer (Vref) on each module.
C6, C7 must sit close to the VM pins: they absorb the motor voltage spikes.

E-stop (on the case)



GPIO9 drives EN_OUT low to enable the drivers.
E-stop pressed or ESP32 in reset: R3 pulls DRV_EN high and both drivers switch off.
GPIO18 reads DRV_EN to detect it.

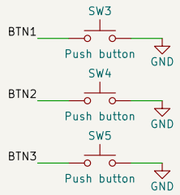
Hand-unit socket (case)



Hand unit (on the cable)



Analog joystick module (GND, VCC, VRx, VRy, SW), powered from 3.3 V.



Buttons and the joystick switch use the ESP32 internal pull-ups.

Notes

1. Supply, controller and drivers are all on the L: no cable crosses the azimuth joint.
2. +12V: 12.0–14.6 V from the battery or a lab supply. It feeds the drivers and U1.
3. TMC2209 header pin order differs between makers: wire by pin name. VIO at 3.3 V (ESP32 logic level).
4. Motor connectors: pins 1–2 one coil, 3–4 the other.
5. A1 pins: 1–22 = header J1, 23–44 = header J3.
6. Hand unit on an 8-wire cable (J2: for example GX16-8, or RJ45 with a network cable).
7. Keep the blue LED dim at night.
8. Board: stripboard (star-track.kicad_pcb). Parts on the case and the hand unit wire to pads P1–P5.

All electronics on the L of the mount
ESP32-S3 controller, two TMC2209 drivers, 12.8 V LiFePO4 battery or lab supply

Sheet: /
File: star-track.kicad_sch

Title: Star-Track mount electronics

Size: A3	Date: 2026-10-26	**Rev: 5**
KiCad E.D.A. 10.0.7		Id: 1/1